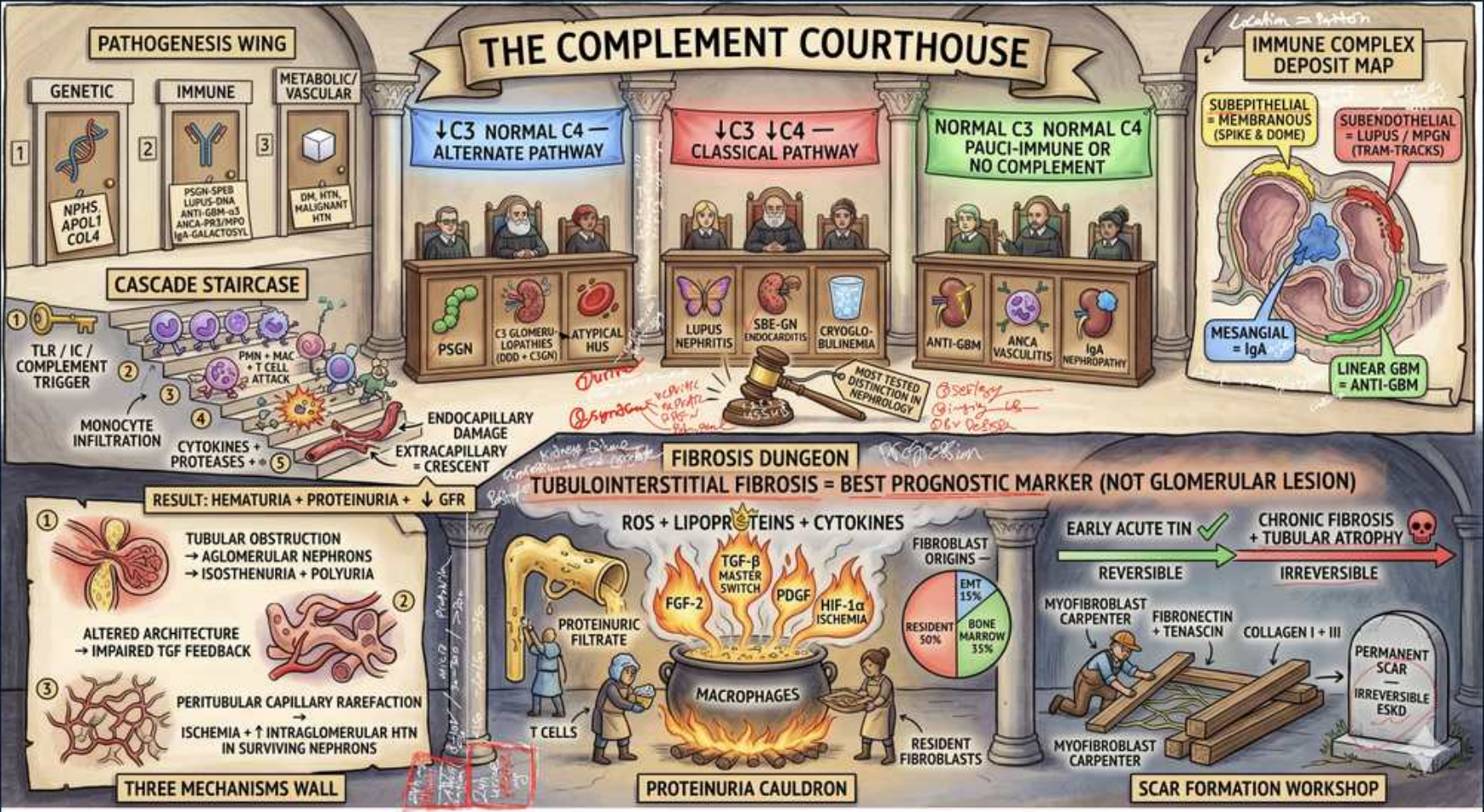


Nephritic Syndrome by Complement — The Complement Courthouse



1. Low C3, Normal C4 → Alternate Pathway

WHAT YOU SEE

Banner, first judge bench

WHAT IT ENCODES

Isolated low C3 with normal C4 implicates alternate-pathway activation. Classic causes: post-streptococcal GN (PSGN) and C3 glomerulopathy/atypical MPGN (C3 nephritic factor). PSGN complement normalizes by 6–8 weeks.

BOARD TRAP

Low C3 + normal C4 narrows to PSGN vs C3 glomerulopathy — persistent low C3 beyond 8 weeks argues against self-limited PSGN and for C3 glomerulopathy.

2. PSGN

WHAT YOU SEE

Judge bench, PSGN label

WHAT IT ENCODES

Post-streptococcal GN follows pharyngitis (1–3 wk) or impetigo (3–6 wk) by nephritogenic strains. Subepithelial 'humps' on EM, granular 'starry sky' IgG/C3 on IF, low C3. Elevated ASO/anti-DNase B. Mostly self-limited in children.

BOARD TRAP

PSGN complement should normalize within ~8 weeks — failure to recover suggests an alternative diagnosis (C3 glomerulopathy, MPGN).

3. Low C3, Low C4 → Classical Pathway

WHAT YOU SEE

Banner, middle bench

WHAT IT ENCODES

Low C3 AND low C4 indicates classical-pathway (immune-complex) consumption. Causes: lupus nephritis, subacute endocarditis-associated GN, and cryoglobulinemia (often hepatitis C). Both complement components are consumed.

BOARD TRAP

Both C3 and C4 low → think immune-complex (lupus/cryo/endocarditis), not the alternate-pathway PSGN/C3 group.

4. Lupus Nephritis

WHAT YOU SEE

Bench, lupus label

WHAT IT ENCODES

SLE GN: 'full house' IF (IgG, IgA, IgM, C3, C1q), low C3 and C4, anti-dsDNA. Class III/IV (proliferative) are most aggressive and need aggressive immunosuppression; class V is membranous/nephrotic. Anti-Sm specific for SLE.

BOARD TRAP

Lupus nephritis class drives therapy — class IV proliferative needs MMF/cyclophosphamide + steroids, while class V membranous is managed differently.

5. Cryoglobulinemia

WHAT YOU SEE

Bench, cryo label

WHAT IT ENCODES

Mixed cryoglobulinemia (type II, often hepatitis C) causes immune-complex MPGN with low C4 (often disproportionately), palpable purpura, arthralgia, and neuropathy. Treat underlying HCV.

BOARD TRAP

Cryoglobulinemic GN classically shows a disproportionately LOW C4 — and HCV is the dominant driver; treat the virus.

6. Normal C3, Normal C4 → Pauci-immune / No Complement

WHAT YOU SEE

Banner, third bench

WHAT IT ENCODES

Normal complement points to pauci-immune (ANCA) vasculitis or anti-GBM disease — little to no immune-complex deposition. These are RPGN causes where complement is typically normal.

BOARD TRAP

Normal complement does NOT exclude severe GN — ANCA vasculitis and anti-GBM are aggressive RPGNs with normal C3/C4.

7. Anti-GBM

WHAT YOU SEE

Bench, anti-GBM label

WHAT IT ENCODES

Anti-glomerular basement membrane disease: linear IgG along GBM on IF, crescents, RPGN; with pulmonary hemorrhage it's Goodpasture syndrome (anti-collagen IV). Normal complement. Treat with plasmapheresis + steroids + cyclophosphamide.

BOARD TRAP

Linear IF = anti-GBM; granular = immune complex; pauci/scant = ANCA — the IF pattern is the key discriminator among RPGNs.

8. ANCA Vasculitis

WHAT YOU SEE

Bench, ANCA label

WHAT IT ENCODES

Pauci-immune crescentic GN: scant immune deposits, normal complement. c-ANCA/PR3 (granulomatosis with polyangiitis), p-ANCA/MPO (microscopic polyangiitis, EGPA). Treat with steroids + rituximab or cyclophosphamide.

BOARD TRAP

ANCA GN shows PAUCI-immune IF (almost no staining) — don't expect the bright deposits of lupus or PSGN.

9. Immune Complex Deposit Map

WHAT YOU SEE

Right panel, deposit diagram

WHAT IT ENCODES

Localization predicts disease: subepithelial humps (PSGN) vs subepithelial spike-and-dome (membranous), subendothelial (lupus/MPGN, 'wire loops'), mesangial (IgA), and linear GBM (anti-GBM). EM/IF location guides diagnosis.

BOARD TRAP

Subepithelial HUMPS (PSGN) vs SPIKES (membranous) are different lesions — humps are nephritic/transient, spikes are nephrotic/chronic.

10. Mesangial IgA

WHAT YOU SEE

Deposit map, mesangial

WHAT IT ENCODES

Mesangial IgA deposition = IgA nephropathy (Berger), the most common GN worldwide; presents with synpharyngitic gross hematuria. IgA vasculitis (HSP) is the systemic form with palpable purpura, arthralgia, abdominal pain.

BOARD TRAP

IgA nephropathy is synpharyngitic (hematuria DURING infection), unlike PSGN which lags 1–3 weeks AFTER — and IgA has normal complement.

11. Linear GBM = Anti-GBM

WHAT YOU SEE

Deposit map, linear pattern

WHAT IT ENCODES

Smooth linear IgG ribbon along the GBM defines anti-GBM disease. Contrasts with the granular 'lumpy-bumpy' pattern of immune-complex GN. Rapidly progressive; plasmapheresis is key.

BOARD TRAP

Linear (anti-GBM) vs granular (immune complex) IF is the single highest-yield distinction on the IF board question.

12. Tubulointerstitial Fibrosis — Best Prognostic Marker

WHAT YOU SEE

Dungeon banner, center

WHAT IT ENCODES

Across glomerular diseases, the degree of tubulointerstitial fibrosis and tubular atrophy on biopsy — not the glomerular lesion itself — best predicts progression to ESRD. Reflects irreversible chronic damage.

BOARD TRAP

Prognosis tracks the INTERSTITIAL fibrosis, not the glomerular activity index — active inflammation can be treated, scar cannot.

13. Three Mechanisms of Progression

WHAT YOU SEE

Wall, lower left

WHAT IT ENCODES

CKD progresses via: (1) tubular obstruction/loss of nephrons → hyperfiltration of survivors; (2) altered glomerular architecture → impaired tubuloglomerular feedback; (3) peritubular capillary rarefaction → chronic ischemia. All converge on fibrosis.

BOARD TRAP

Surviving-nephron hyperfiltration is adaptive short-term but maladaptive long-term — the rationale for RAAS blockade to lower intraglomerular pressure.

14. Proteinuria Cauldron

WHAT YOU SEE

Cauldron, macrophages/cytokines

WHAT IT ENCODES

Filtered proteins are themselves nephrotoxic: they activate tubular cells to release ROS, lipoproteins, and cytokines, recruiting macrophages and T cells. This profibrotic cascade explains why proteinuria magnitude predicts progression and why lowering it (RAAS blockade) is protective.

BOARD TRAP

Proteinuria is not just a marker — it actively drives interstitial injury, which is why reducing proteinuria is a treatment target, not only a metric.

15. Myofibroblast / Scar Formation

WHAT YOU SEE

Scar workshop, collagen carpenters

WHAT IT ENCODES

TGF-beta and PDGF drive fibroblast-to-myofibroblast transformation, depositing fibronectin, tenascin, and collagen I/III. Early TIN is reversible; established collagen scar (tubular atrophy) is permanent ESKD.

BOARD TRAP

The reversible/irreversible line is TGF-beta-driven collagen scar — once myofibroblasts lay down collagen I, the damage is fixed.